1 Introduction

This project studies portfolio-level clustering for U.S. small-cap equities. We compare three complementary approaches—(i) Path-Signature features with K-means, (ii) correlation-based hierarchical clustering, and (iii) Dynamic Time Warping (DTW) on cumulative log returns—to uncover structure in small-cap return dynamics. Unlike single-name idiosyncrasies or one-shot cross-sectional snapshots, our focus is on rolling time windows, allowing us to track how clusters form, split, and migrate across regimes (e.g., risk-on vs. risk-off, liquidity squeezes, rate shocks). The end goal is to deliver clusters that are both statistically robust and financially interpretable—for instance, distinguishing quality vs. junk, commodity-linked names, or growth/long-duration profiles inside the small-cap universe.

2 Motivation

Small caps are an attractive but messy corner of the market. They exhibit higher volatility, weaker analyst coverage, thinner liquidity, and sharper microstructure noise than large caps. Classical correlation clustering often collapses these effects into one factor (the "market mode"), missing path-shape information such as "fast crash then rebound," volatility timing, and asymmetric drawdowns that practitioners care about. Path-Signature features explicitly encode the trajectory of returns (and auxiliary channels like realized volatility), offering a richer representation than correlation alone. DTW adds a shape-alignment baseline to test whether timing differences—not just co-movement—drive the grouping.

From a portfolio-construction standpoint, small-cap clustering can:

- Improve diversification by avoiding look-alike exposures that standard sector tags fail to capture.
- Support regime-aware rebalancing, as clusters reconfigure under liquidity stress or rate cycles.
- Enhance risk diagnostics, surfacing latent groups (e.g., high-beta and low-profitability "junk" pockets) that inflate tail risk.

In short, we are motivated to test whether Signature-based clustering delivers more stable, interpretable, and actionable groupings in small caps than traditional correlation or DTW methods—especially when evaluated through rolling windows, stability metrics, and simple economic diagnostics (e.g., factor loadings, drawdown profiles).

3 Data

Our analysis combines two datasets: CRSP security-level equities data and an equity news sentiment feed from Alexandria Technology.

The core panel is drawn from the Center for Research in Security Prices (CRSP). We use CRSP's security identifiers (PERMNO/PERMCO), ticker, company name, exchange code (EXCHCD), security code (SHRCD), four-digit SIC industry code (SICCD), price (PRC), shares outstanding (SHROUT), and daily simple return (RET). This panel allows us to define a point-in-time universe of U.S. common stocks (NYSE/AMEX/NASDAQ) in a target market-capitalization range of 2–10 billion USD from 2015 to 2025, then observe those same securities at daily frequency within the study windows. CRSP is accessed through Wharton Research Data Services (WRDS); replication simply requires querying the same fields for the same anchor dates and windows and recording the WRDS pull date.

For news-based annotations we use Alexandria Technology's Equity Newswires feed. The vendor delivers time-stamped, security-linked news records with fields such as headline/source identifiers, ticker and ISIN, directional sentiment scores with model confidence, novelty/relevance flags, and event tags. This feed is obtained directly from Alexandria under license; replicators should request the same product and historical window and note the vendor delivery date and schema version.

4 Methodology

4.1 Data Preprocessing

The raw CRSP dataset contains daily price (`PRC`) and share count (`SHROUT`) observations for all listed securities. To ensure completeness and consistency, only stocks with at least 2,500 trading-day observations were retained. Market capitalization was computed as

$$\text{MarketCap}_{i,t} = |\text{SHROUT}_{i,t} \times \text{PRC}_{i,t} \times 1000|.$$

We selected securities whose most recent market capitalization lay between \$2 billion and \$10 billion, corresponding to the conventional definition of the small-cap segment. Only common shares listed on major U.S. exchanges (NYSE, AMEX, NASDAQ) were included.

To standardize the scale of returns, we converted price series into continuously compounded daily log returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right),$$

which are additive over time and more suitable for statistical modeling. Stocks with non-positive prices or missing values were removed. Finally, we reshaped the data into a wide-format matrix $\mathbf{R} \in \mathbb{R}^{T \times N}$, where each column represents a stock's log return time series and each row corresponds to a trading day.

4.2 Clustering Algorithms

4.2.1 Ledoit–Wolf Correlation-Based Clustering

The first clustering approach relies on the linear dependence structure of asset returns. Let $R \in \mathbb{R}^{T \times N}$ denote the matrix of daily returns, where T is the number of time periods and N is the number of assets. To obtain a stable estimate of the covariance matrix in the presence of limited observations and high dimensionality, we use the Ledoit–Wolf shrinkage estimator as implemented in the `scikit-learn` library. This estimator automatically determines an optimal shrinkage intensity and returns a regularized covariance matrix $\hat{\Sigma}_{LW}$, which reduces estimation noise while retaining essential correlation structure.

The corresponding correlation matrix is computed as

$$\rho_{ij} = \frac{\hat{\Sigma}_{LW,ij}}{\sqrt{\hat{\Sigma}_{LW,ii}\hat{\Sigma}_{LW,jj}}},$$

where ρ_{ij} measures the degree of linear co-movement between assets i and j. The correlation matrix is then transformed into a distance matrix

$$D_{ij} = \sqrt{2(1 - \rho_{ij})},$$

so that more highly correlated assets are closer in distance space. Hierarchical agglomerative clustering with Ward linkage is then applied to D to form groups of assets that exhibit similar correlation structures. This method captures relationships driven primarily by linear dependencies in return dynamics and serves as a baseline for comparison with nonlinear approaches.

4.2.2 Signature-Based Clustering

The second clustering framework leverages the rough path *signature* representation to capture nonlinear and temporal characteristics of asset return trajectories. Unlike correlation-based methods, which summarize linear co-movements, this approach embeds the entire path evolution of each asset into a high-dimensional feature space derived from iterated integrals.

Each asset's cumulative return path is first constructed as

$$P_t^i = \prod_{\tau \leq t}(1 + r_\tau^i),$$

and normalized to start at one to ensure comparability across assets. To preserve the temporal ordering of price movements, we apply a *lead–lag transformation*, which maps each one-dimensional time series into a two-dimensional path that encodes both the current and lagged values. This transformation enhances the representation of sequential structure, making it suitable for computing path signatures.

We then compute the truncated path signature of each transformed trajectory using the `iisignature` library. For a continuous path $X : [0, T] \to \mathbb{R}^d$, the signature of order m is defined as the sequence of all iterated integrals:

$$S^{(m)}(X) = \left(1, \int dX_t, \int dX_{t_1} \otimes dX_{t_2}, \dots, \int_{0<t_1<\cdots<t_m<T} dX_{t_1} \otimes \cdots \otimes dX_{t_m}\right).$$

In our implementation, we truncate the expansion at depth $m = 4$, balancing representational richness and computational tractability. Each signature is then flattened into a feature vector, forming a feature matrix

$$\mathbf{S} = [S^{(4)}(X^1), S^{(4)}(X^2), \dots, S^{(4)}(X^N)]^\top,$$

where each row corresponds to one asset.

We perform agglomerative hierarchical clustering with Ward linkage directly on the Euclidean distances in this signature feature space. This groups assets whose return trajectories exhibit similar temporal shapes and nonlinear behaviors, even if their linear correlations differ. Visualization through t-SNE projection of the signature features provides geometric insight into the structure of these clusters, while cumulative return analysis reveals how each cluster evolves over time.

4.3 Cluster Selection and Rolling-Window Analysis

4.3.1 Optimal Number of Clusters.

Because the correlation-based and signature-based approaches encode different types of information, we employ distinct criteria to determine the optimal number of clusters in each case.

For the *Ledoit–Wolf correlation-based method*, the number of clusters k is chosen by maximizing the *silhouette score*, which evaluates how well each observation fits within its assigned cluster relative to others. The silhouette coefficient for an observation i is defined as

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))},$$

where $a(i)$ is the average distance between i and other points in the same cluster, and $b(i)$ is the minimum average distance between i and all other clusters. We compute the average silhouette score for $k = 2, \dots, 10$ and select the k that maximizes it, yielding geometrically compact and well-separated clusters. This metric is particularly suited to the Euclidean geometry of the correlation feature space.

In contrast, for the *path-signature clustering method*, we select k by maximizing the *Maximum Mean Discrepancy* (MMD) between cluster distributions. Given two clusters X and Y, MMD measures the distance between their probability distributions under a reproducing kernel Hilbert space:

$$\mathrm{MMD}^2(X, Y) = \mathbb{E}[k(x, x')] + \mathbb{E}[k(y, y')] - 2\,\mathbb{E}[k(x, y)],$$

where $k(\cdot,\cdot)$ denotes the RBF kernel. We compute all pairwise MMD values among clusters and take their average. The optimal k is the one that maximizes the average MMD, corresponding to the configuration that yields the most distinct feature distributions. This kernel-based approach is more appropriate for nonlinear representations such as path-signature embeddings.

4.3.2 Rolling-Window Framework.

To capture the evolution of clustering structure over time, we apply a rolling-window analysis. We partition the ten-year dataset into overlapping windows of five years (approximately 1,260 trading days), moving forward by one year (252 trading days) at each step. Within each window, we re-estimate features, determine the optimal number of clusters using the corresponding criterion (silhouette or MMD), and perform hierarchical agglomerative clustering. This procedure produces a sequence of time-varying cluster assignments that reflect changing market regimes and cross-sectional dependencies.

4.4 Visualization and Interpretation.

For each rolling window, we visualize the results using several diagnostic plots:

- A clustered heatmap of the correlation matrix, displaying visually distinct blocks of co-moving assets.
- Two-dimensional *t-SNE* embeddings of the path-signature features, providing a geometric view of cluster separation in nonlinear space.
- Cumulative return plots of all tickers within each cluster, showing both individual trajectories (semi-transparent) and the cluster's mean return path.

These visualizations allow qualitative assessment of how clusters evolve and how stable each grouping remains across time windows.

4.4.1 Equal-Weighted Cluster Portfolios.

Within each window, we plot equal-weighted portfolio returns of all clusters to qualitatively evaluate clustering effectiveness. Given cluster c containing N_c assets, the cluster-level return on day t is defined as

$$R_{c,t} = \frac{1}{N_c} \sum_{i \in C_c} r_{i,t},$$

and its cumulative return is obtained as the compounded product of $(1 + R_{c,t})$ over the window. Distinctly separated return trajectories indicate that the clustering successfully groups assets with similar performance dynamics.

4.4.2 Performance Tracking Across Windows.

The rolling-window design further allows analysis of cluster persistence and regime shifts. We track each cluster's cumulative performance, volatility, and Sharpe ratio over time to evaluate both the stability and predictive power of the clustering scheme. This dynamic structure highlights whether certain economic regimes (e.g., risk-on or risk-off) favor specific cluster types—such as high-volatility "junk" clusters versus stable "quality" ones—providing insight into the time-varying behavior of small-cap equity groups.

4.5 Post-Clustering Evaluation and Diagnostics

4.5.1 Feature-Based Cluster Characterization.

To assess the statistical and financial properties of each cluster, we compute a set of descriptive features for every ticker within each rolling window. For each asset's log-return series $r_{i,t}$, we derive the following metrics:

- **Drift**: annualized mean return $\mu_i = 252 \times \bar{r}_i$,
- **Volatility**: annualized standard deviation $\sigma_i = \sqrt{252}\,\mathrm{sd}(r_i)$,
- **Skewness** and **Kurtosis**: higher moments of the return distribution capturing asymmetry and tail heaviness,
- **AC1**: first-order autocorrelation coefficient, $\rho_i(1)$, measuring short-term return persistence,
- **Maximum Drawdown (MDD)**: largest cumulative loss from a peak in the cumulative wealth process

$$\mathrm{MDD}_i = \min_t \frac{W_t}{\max_{\tau \le t} W_\tau} - 1, \quad \text{where} \quad W_t = \prod_{\tau \le t} (1 + r_{i,\tau}).$$

Within each window, these quantities are averaged across all assets in the same cluster, yielding a table of cluster-level descriptive statistics:

$$\mathrm{ClusterFeature}_c = \frac{1}{N_c} \sum_{i \in C_c} \left[\mu_i, \sigma_i, \mathrm{Skew}_i, \mathrm{Kurt}_i, \rho_i(1), \mathrm{MDD}_i\right].$$

Scatterplots such as Drift–Volatility, Skewness–Kurtosis, and Volatility–Drawdown are used to visualize the relationships among these metrics and to interpret the economic characteristics of each cluster (e.g., "high-drift/high-volatility" vs. "low-volatility/quality" clusters). In each plot, the points are colored according to their cluster assignments. When the points belonging to different clusters form visibly distinct and non-overlapping regions, it suggests that the clustering has successfully captured meaningful heterogeneity in return behavior rather than random variation. Conversely, substantial overlap across colors may indicate weak cluster separation or unstable feature relationships within the given window.

4.5.2 Cluster Compactness and Separation.

To evaluate the internal consistency of clusters and the degree of distinction among them, we compute two complementary quantities for each window:

$$\text{Compactness} = \frac{1}{|C|}\sum_{c\in C}\frac{1}{|P_c|}\sum_{(i,j)\in P_c} d(x_i, x_j),$$
$$\text{Separation} = \frac{1}{|Q|}\sum_{(c_1,c_2)\in Q}\frac{1}{|C_{1,2}|}\sum_{i\in C_{c_1}}\sum_{j\in C_{c_2}} d(x_i, x_j),$$

where $d(\cdot,\cdot)$ denotes pairwise Euclidean distance, P_c is the set of within-cluster pairs, and Q enumerates all distinct cluster pairs. We also track the ratio Separation/Compactness as a summary measure of cluster quality. Trends in these quantities across rolling windows reveal whether clustering structure strengthens or weakens over time.

4.5.3 Transition Matrices and Cluster Persistence.

To quantify the stability of cluster memberships across consecutive rolling windows, we construct *transition matrices*. For each adjacent pair of windows, we map each stock's cluster assignment $(C_i^{(t)}, C_i^{(t+1)})$ and compute transition counts:

$$M_{ab} = |\{i : C_i^{(t)} = a,\, C_i^{(t+1)} = b\}|,$$

which are then row-normalized to represent percentage transitions from cluster a at time t to cluster b at time $t+1$. Visualizing these matrices as heatmaps highlights both persistent clusters (dominant diagonal) and reallocation dynamics (off-diagonal mass). Stable diagonal patterns indicate consistent behavior types, while more diffuse matrices suggest regime shifts or increased volatility in the clustering structure.

4.5.4 Interpretation.

Together, these diagnostics provide a multifaceted view of cluster behavior: feature-based summaries capture within-window statistical structure, compactness and separation evaluate the geometric quality of the clustering, and transition matrices track temporal stability. These analyses collectively allow us to determine whether clusters are economically interpretable, statistically robust, and persistent across changing market conditions.

5 Empirical Results

5.1 Clustering Visualization and Cumulative Returns.

Having applied both the Ledoit–Wolf correlation-based clustering and the path-signature-based clustering frameworks across rolling windows, we present here the results for the most recent window (2019–2024) as a representative example. This period captures multiple market regimes—from the 2020 COVID drawdown to the subsequent recovery—and thus serves as a robust test of clustering stability and interpretability. Both methods are evaluated using visual diagnostics, portfolio-level performance, and feature-based characterization.

5.1.1 Ledoit–Wolf Correlation-Based Clustering.

The Ledoit–Wolf estimator regularizes the sample covariance matrix to stabilize correlation estimates in high-dimensional settings. Figure 1 presents the Ledoit–Wolf correlation matrix (left) and the cumulative return paths of the resulting cluster portfolios (right). The block-diagonal structure in the heatmap reveals two large, distinctly separated groups of assets with strong intra-cluster correlations and weak cross-cluster links. This outcome suggests that correlation-based clustering successfully captures the dominant linear market structure, segmenting the universe into broad, economically interpretable clusters.

The portfolio plot on the right further supports this view: the two equal-weighted cluster portfolios exhibit persistently divergent return trajectories, implying that the Ledoit–Wolf clusters correspond to meaningfully different economic exposures—such as cyclical versus defensive stocks—rather than random partitions.

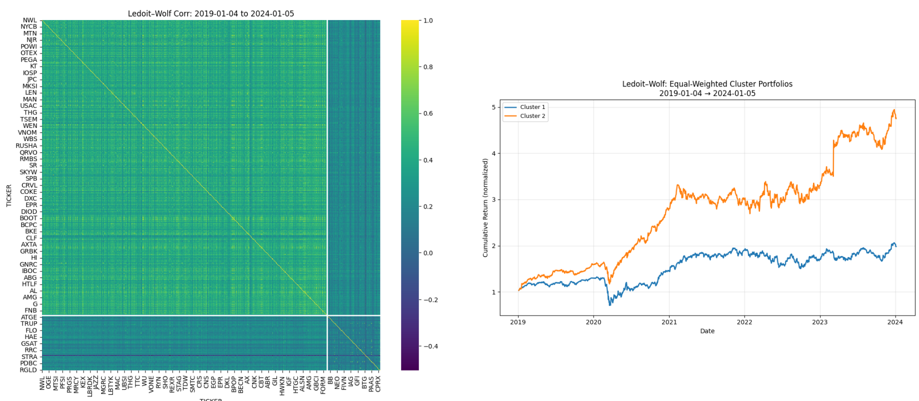

Figure 1: Ledoit–Wolf correlation-based clustering (2019–2024). **Left:** Ledoit–Wolf shrinkage correlation matrix with block-diagonal structure. **Right:** Equal-weighted cluster portfolios. Distinct performance paths indicate meaningful economic segmentation.

5.1.2 Path-Signature-Based Clustering.

In contrast, the path-signature method captures finer-grained structure by embedding each normalized price trajectory into a high-dimensional space that encodes its temporal evolution. Figure 2 illustrates the t-SNE projection of signature features (left) and the corresponding equal-weighted cluster portfolio performance (right). Unlike the broader separation seen in the Ledoit–Wolf case, the signature-based embedding reveals a more granular arrangement—multiple compact clusters that remain visibly distinct but closer in the feature space. This pattern reflects the method's sensitivity to subtle nonlinear characteristics such as volatility timing, recovery shape, and path asymmetry.

The corresponding portfolio plot indicates that these clusters differ not only statistically but also economically, though the separation occurs along more nuanced behavioral dimensions rather than large macro-style splits. This highlights the method's ability to uncover second-order heterogeneity—variations in trajectory shapes even among assets that move together on average.

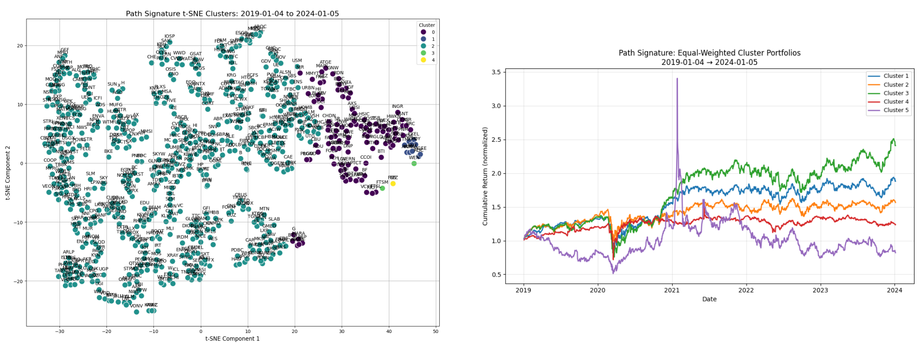

Figure 2: Path-signature-based clustering (2019–2024). **Left:** t-SNE projection of signature features with clear color-separated clusters. **Right:** Equal-weighted cluster portfolios showing stronger dispersion across clusters, reflecting nonlinear trajectory segmentation.

5.2 Signature-Based Cluster Characterization (2016–2021 Window).

To evaluate how effectively the signature based clustering method differentiates assets based on their statistical behavior, we examine the feature distribution across clusters in the 2016–2021 window. Compared with later periods, this window exhibits a more balanced cluster composition, allowing clearer inspection of whether distinct risk–return characteristics, tail behaviors, and temporal properties are meaningfully separated.

Risk–Return Structure. Figure 3 plots annualized drift versus volatility. Unlike the later window where one large cluster dominates, the 2016–2021 period shows ten clusters with a more even spread. The clustering algorithm achieves clear differentiation between moderate-risk and high-volatility stocks: low-drift, high-volatility assets (typically riskier or more cyclical names) occupy distinct peripheral clusters, while stable, mid-volatility stocks form compact central groups. This balanced dispersion suggests that the clustering method captures genuine heterogeneity in risk exposure, rather than over-collapsing assets into a single market factor.

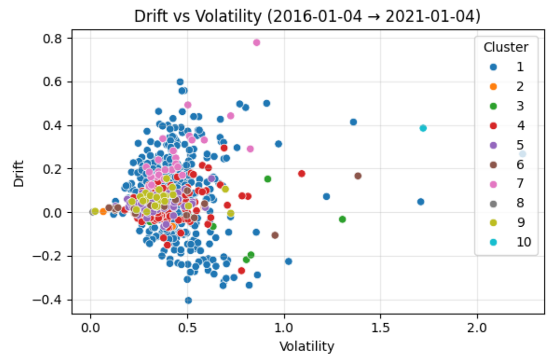

Figure 3: Drift vs. Volatility (2016–2021). More even cluster allocation highlights separation along risk–return dimensions, isolating high-volatility and low-return stocks from the central mass of moderate-risk assets.

Higher-Moment Differentiation. Figure 4 examines skewness and kurtosis, which capture asymmetry and tail heaviness in returns. The plot reveals that the clustering algorithm distinguishes between assets with heavy-tailed and near-normal return distributions: clusters positioned along the extreme ends of the skewness axis correspond to stocks with large positive or negative jumps, while centrally located clusters group assets with balanced return profiles. The presence of several small but compact clusters around extreme kurtosis values demonstrates that the clustering successfully identifies outliers in tail behavior rather than treating them as noise.

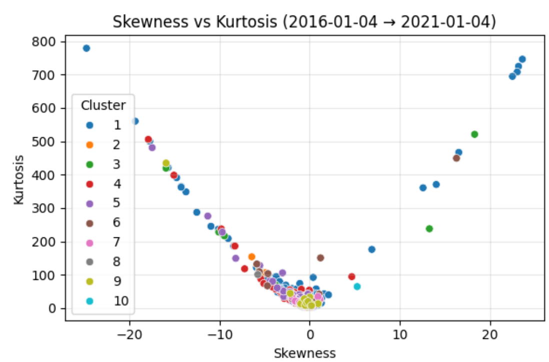

Figure 4: Skewness vs. Kurtosis (2016–2021). Clusters occupy distinct regions of the tail-risk spectrum, showing the method's ability to isolate non-Gaussian return behavior.

Temporal Dependence Separation. Figure 5 shows the relationship between one-lag autocorrelation (AC1) and drift. Here, clustering appears effective at distinguishing stocks by temporal dependence: several clusters contain assets with positive or near-zero AC1 (mild trend persistence), while others aggregate negatively autocorrelated (mean-reverting) names. This spread across the AC1 dimension, coupled with distinct coloring patterns, indicates that the clustering captures dynamic properties of returns—beyond static volatility levels—suggesting sensitivity to time-series structure.

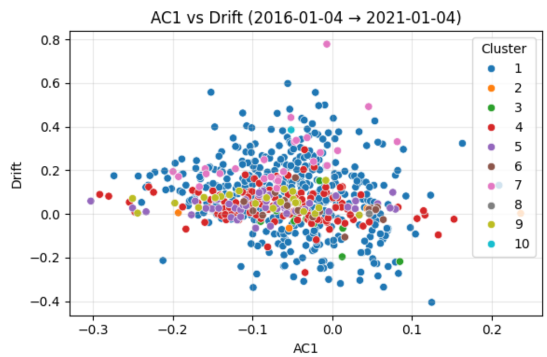

Figure 5: AC1 vs. Drift (2016–2021). The method achieves visible separation along autocorrelation, distinguishing trend-following versus mean-reverting behaviors.

Downside Risk Segmentation. Finally, Figure 6 compares volatility and maximum drawdown. The negative slope between the two variables remains strong, but now clusters are more evenly distributed along this frontier. Certain clusters clearly represent low-volatility, shallow-drawdown names, while others consist of highly volatile assets that experienced extreme losses. This separation suggests that the clustering algorithm meaningfully partitions stocks by downside risk intensity, yielding clusters that align with different levels of vulnerability to market shocks.

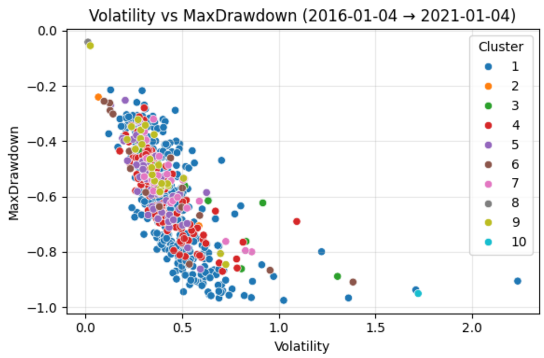

Figure 6: Volatility vs. Maximum Drawdown (2016–2021). Clusters segment the volatility–drawdown frontier, separating stable from high-risk assets with deeper cumulative losses.

5.3 Cluster Transition and Temporal Stability.

To evaluate the persistence of clustering structures across rolling windows, we compute transition matrices that track how assets migrate between clusters over time. Each matrix compares two consecutive windows, where rows represent cluster membership in the earlier period and columns represent cluster membership in the subsequent period. Values are normalized by row, indicating the percentage of assets that remained within or moved between clusters. A highly stable clustering would exhibit strong diagonal dominance (most assets remaining in the same cluster), while greater off-diagonal dispersion suggests reorganization of the market structure.

Path-Signature Clustering. Figure 7 presents the transition matrix for the path-signature–based clustering between the 2018–2023 and 2019–2024 windows. The result displays a moderate degree of persistence: the largest group (`C0`) retains 99.4% of its members, indicating a highly stable core cluster with consistent trajectory-level characteristics. However, the secondary cluster (`C1`) shows that 42% of its constituents migrated into `C0` in the following window, suggesting partial convergence of patterns previously considered distinct. This asymmetry implies that while the main structure of the market remained stable, certain previously separate subgroups lost distinctiveness—potentially reflecting a regime in which path-shape heterogeneity decreased as macro factors became dominant.

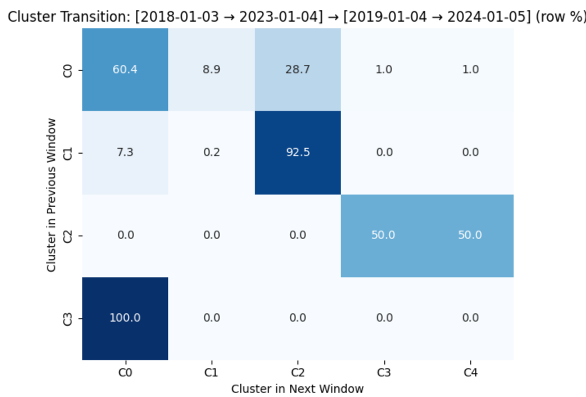

Figure 7: Cluster transition matrix for path-signature clustering between the 2018–2023 and 2019–2024 windows (row-normalized percentages). Diagonal dominance reflects strong persistence for the main cluster, while off-diagonal migration captures partial convergence of smaller groups.

Correlation-Based Clustering. Figure 8 shows the transition matrix for correlation-based clustering over the same pair of windows. Here, the transition patterns are more fragmented: several clusters (`C0`, `C2`, and `C3`) display substantial cross-flows, with 28–40% of assets reassigning to other groups. For instance, `C0` disperses across multiple destinations (28.7% shifting into `C2`), and `C2` itself splits evenly between two next-period clusters. Only `C1` shows high persistence (92.5%), corresponding to a subset of stocks whose correlation profiles remain stable across time. This result suggests that correlation-based clustering is more sensitive to shifts in co-movement patterns—potentially capturing cyclical rotations or changes in sector linkages—but at the cost of temporal consistency.

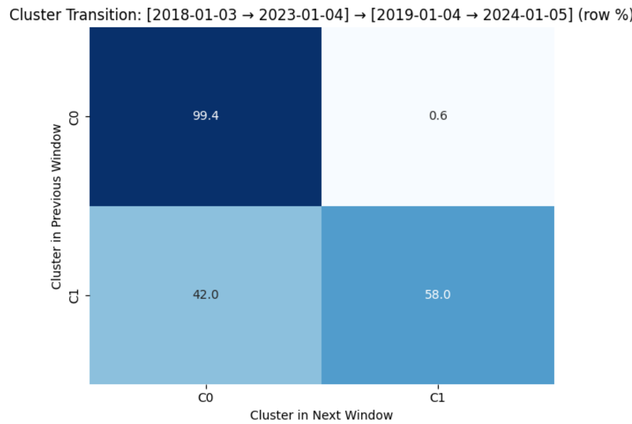

Figure 8: Cluster transition matrix for correlation-based clustering (2018–2023 → 2019–2024). Multiple off-diagonal flows reveal greater reorganization of co-movement structures relative to the path-signature method.